



SKULLBONEZ

CORE

FIRST-TIME USER MANUAL

Explore. Build a scene. Test what happens.

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Your first session

This guide helps you explore, build and test your own SkullbonezCore scenes.

1. Extract, then launch

Download the Windows ZIP from the [public release page](#). Extract the entire archive to a writable folder and run **Launch SkullbonezCore.cmd**. Keep the executable, DLLs and **SkullbonezData** together. You need 64-bit Windows and a DirectX 12-compatible GPU and driver. This build is unsigned; no development tools are needed.

2. Learn to look and move

Choose **Start tutorial** on first launch, or click the top-bar **Tutorial book**. The tutorial minus button collapses it; plus or the book restores it. The first lesson teaches the camera. Hold **right mouse** over the scene and move the mouse to look around. Keep it held and use **W/A/S/D** to move. Release right mouse to use panels and tools again.

3. Find the controls

Open **Options** at the bottom: scene tools are on the left, diagnostics on the right. **Scene** loads levels; **Keys & Help** lists controls. **Video** on the right holds graphics settings. The header shows the level and **Edit mode / Playback**. Hover for help; scroll inside panels.

Start with these controls

Control	What it does
Right mouse + move mouse	Look around the viewport. In an orthographic view, pan instead.
Right mouse + W/A/S/D	Move forward, left, backward and right. Q/E moves down/up.
Shift / wheel while looking	Temporary speed boost / adjust flight speed.
Middle-drag / idle wheel	Pan the camera / zoom.
Double-click an object	Attach the camera. For a car, press T to enter Drive.
W/A/S/D / Space in Drive	Accelerate, steer, reverse / brake. Esc detaches.
Top-bar Edit mode / Playback	Red means editing; green means playback. Click to switch. Edit scene mirrors it.
0 (zero)	Show or hide the UI.

Good to know

The tutorial covers driving, rewinding, editing, terrain and the Solar System. **Retry section** restores the current chapter; **Skip step** moves past a lesson. Practice scenes are protected: use **New scene** to make a level you can save.

Adjust the workspace and look

Drag the grip between Editor and Replay/Camera to change their heights. **Options > Video** offers **Max / Medium / Min** presets and live FPS. Changes last this run unless you click **Save settings**. In **Editor > Material**, select an object and choose a **Finish**; adjust **Roughness** and **Reflectivity**. **Ctrl+Z** undoes a material change; **Save scene** keeps it.



Build your first scene

Start with a box and ball on flat terrain. Objects also includes **Domino** and **Red arch**.

1. Open the Editor

Click **Editor** > **Controls** to enter editing. Save existing work before starting a new level.

2. Create a new scene

Click **New scene** for a blank editable scene. Its name appears in **Options** > **Scene**.

3. Find the placement controls

Open **Objects** and choose a type from the scrollable object list to enter placement automatically. **Edit scene** controls editing; **Select [Q]** returns to selecting existing objects.

4. Place a box

Choose **Box**. Turn **Static object** on for a fixed obstacle, or off for a body that can fall. Move onto terrain until the preview appears, then click and release left mouse to place it.

5. Add a ball

Choose **Ball** and turn **Static object** off. Move the preview near the box. Before placing, use the wheel to raise it above the terrain, then click and release. Page 3 explains sizing and rotation.

6. Save the scene

Click **Save scene** in **Editor** > **Controls**. This writes the authored level under **SkullbonezData/scenes/**. Check the save feedback before changing scenes or closing the app. **F2** makes a separate scene shot; it does not save the active level.

7. Run your experiment

Click **Play**, including from Edit mode. Press **6** to restore your edited starting arrangement and clear the timeline. Unsaved edits survive reset; reset does not save the scene.

8. Reopen and try again

Find saved levels in **Options** > **Scene**. Edit, save and run again. Scenes and replay recordings are separate saves (page 4).

Good to know

Creating a scene does not replace saving it. Use **Save scene** for your authored level. Tutorial practice scenes cannot be overwritten; create your own level to keep changes.

Minecraft mouse controls

Enable **Minecraft mode** in Controls. Click the viewport to capture the mouse; **left-click places**, **right-click removes**, **Esc releases**. **Wheel zooms**, **Ctrl-wheel rotates**, **Shift-wheel adjusts height**. **1 Move / 2 Rotate / 3 Scale** release the mouse for editing; **4 turns 90 degrees / 5 snaps to ground / 6 resets**. Keys 4/5 affect the preview or selected blocks. Choose an object in Objects to resume placing.

Ragdoll colliders

Open **Physics** > **Visualisation** > **Ragdoll colliders: Off, Selected Ragdolls**, or **All Ragdolls**. Select any part to see its whole ragdoll. New heads use sphere colliders; older saved box heads retain their physics.



Place, select, transform

The pointer acts on the viewport. A click over a panel operates that panel instead.

Mode	Use it for
Choose a type in the object list	Preview and insert a new object. Placement enters Edit mode automatically.
Select [Q]	Select an existing object, then choose Move, Rotate or Scale. A gizmo is the visible set of transform handles.

Placing objects

Gesture or control	Result
Left click and release	Place the current preview. Placement commits on release.
Hold left mouse + drag	Resize before placing. Boxes use drag for footprint dimensions and wheel for height during the drag; round objects scale uniformly.
Wheel before clicking	Raise/lower the preview; use Shift-wheel in Minecraft, where plain wheel zooms.
Ctrl + wheel	Rotate the preview around the vertical axis in 15-degree steps.
Static object / Terrain align	Choose fixed or dynamic physics / align placement to the terrain slope.

Editing existing objects

Click **Select [Q]**, then an object. Keep **Modify velocity** off for shape edits. Release right mouse before using tool shortcuts.

Tool or gesture	Result
Move [W]	Drag an arrow along one axis, or a square handle in a plane.
Rotate [E]	Drag a ring to rotate around its axis.
Scale [R]	Drag the centre for uniform size, or an axis for supported shape dimensions.
Space: World / Local	Choose fixed world axes or axes aligned to the object.
Move / Angle / Scale snap	Toggle snapping and choose each increment with its value control.
Esc during a drag	Cancel the edit and restore the starting transform.
Ctrl-click / Ctrl+C / Ctrl+V	Add to a selection / copy / paste standalone objects together, keeping their arrangement.
Make static / Make physical	Convert selected blocks in place, including blocks in an older saved level. Save afterwards.
Delete / Ctrl+Z / Ctrl+Y	Delete / undo / redo. Ctrl+Shift+Z also redoes.

Camera: right mouse + W/A/S/D flies; middle-drag pans. **Frame selected [F]** fits the selection in view.

Good to know

Save your starting scene. Use undo before leaving Edit mode, which clears edit history. A grey axis means that change is unavailable.



Physics, prediction & replay

Start simple: save a scene, run it, inspect what happens, then reload your saved setup.

Control the simulation

Use **Play/Pause** on the bottom transport; Play can leave Edit mode for you. The scrubber is grey and cannot be dragged until there is recorded or predicted time to inspect. Play creates a new future. Static bodies stay fixed; physical bodies can move.

1. Predict and inspect

Open **Replay** and click the bold **Predict** control, then click a body in the viewport. Wait for its future path and **Causes**. Click a cause to inspect a collision; **Exit inspect** or **Esc** leaves inspection without launching the paused experiment.

2. Compare a different future

Modify velocity sits directly below Predict. With a predicted object selected, enable it, drag a velocity arrow and release. Compare **Original** and **Modified** by scrubbing the timeline. Choose **Accept Original** or **Accept Modified**, turn Predict off, then press **Play** to launch it.

3. Rewind, edit or branch

Drag the timeline thumb to inspect recorded history. Switch to **Edit** to return to the starting arrangement. If you change nothing, **Playback** returns to the retained time. The first edit discards that future, even if undone; Play then runs from the start. **Branch from here [Enter]** in Replay instead resumes from the selected recorded point.

Understand the different saves

Action	What you keep
Editor > Controls > Save scene	The editable authored level under SkullbonezData/scenes/.
F2	A separate numbered scene shot; it does not save the active level.
F3	A viewport screenshot.
Replay > Save / Load recording	Recorded simulation data, separate from the editable scene. High detail prediction is just above Save recording.

Good to know

Reset [6] restores the edited starting arrangement and clears the timeline. Unsaved editor changes survive reset. **Reset Defaults** reloads authored defaults; save before using it or changing scenes.

Director camera

In Edit mode, **Camera** replaces Replay in the lower dock. Move with right mouse and W/A/S/D; release it and **Capture view** at two or more positions. Choose **Curve**, easing and **Faster / Slower**, then **Play path**. **Space** pauses; hold **B** to grab the view. Backtick returns to Edit. **Save path / Load path** use a separate `.shot.json` file.

Explore further

For a space experiment, load **solar_system.scene.json**, set **Prediction horizon** to **40s**, select **Earth**, **Predict**, then **Play**.

Solver Lab requires separately supplied comparisons; the portable ZIP omits them. Space plays/pauses; left/right arrows step.



Keyboard & mouse

Finish typing or close popups before using shortcuts. Release right mouse for tools. Vehicle and launcher controls depend on their modes.

Key / gesture	Action and context
Right mouse + move mouse	Viewport camera look; pan in an orthographic view.
Right mouse + W/A/S/D; Q/E	Fly forward/left/back/right; down/up.
Shift / wheel while looking	Temporary faster travel / adjust flight speed.
Middle-drag / idle wheel	Pan / zoom.
Alt-left / Alt-right / Alt-middle drag	Orbit / dolly / pan around the current focus.
Q / W / E / R	Select / Move / Rotate / Scale in editing and free inspection.
F	Frame the selection in editing and free inspection.
Backtick (`)	Enter or leave Edit mode (US keyboard key below Esc).
Double-click / Esc	Attach to an object / detach. Esc dismisses or cancels an active interaction first.
T	Enter or leave Drive when attached to a car.
W/A/S/D / Space in Drive	Accelerate, steer, reverse / brake.
Ctrl+C / Ctrl+V	Copy / paste the selected standalone objects, including a multiple selection.
Ctrl+Z / Ctrl+Y or Ctrl+Shift+Z	Undo / redo an editor edit.
Delete / Esc during a transform	Delete the editable selection / cancel the drag.
Ctrl + wheel in placement	Rotate the preview by 15 degrees per wheel step.
P	Toggle paused prediction inspection. Use visible Play to run afterwards.
F1 / Enter when attached	Cycle attached-camera submode / toggle camera pin.
Enter in replay	Branch from the selected recorded point when available.
F2 / F3	Save a separate scene shot / screenshot.
0 / F7	Show/hide UI / switch Split Future and legacy lighting.
N / M	Toggle launcher / cycle its fire mode while active.
Space / left-right in Solver Lab	Play/pause / step backward-forward.
V / C / G / O	Collision visuals / physics overlays / broadphase bounds / terrain contact probe.
1 / 2 / 3 in Minecraft	Move / Rotate / Scale; replaces the old box-size presets.
4 / 5 in Minecraft; 6 / Backspace	Turn 90 degrees / snap to ground; reset scene and clear timeline.

Q/W/E/R still select tools in Minecraft after **Esc** releases mouse-look; captured **WASD** moves the camera. More controls: **Options > Keys & Help** and hover help.



When something feels stuck

Most first-session surprises come from an active mode, a paused scene, or a pointer over a panel.

What you see	What to check
Objects will not move	Turn Predict off and press Play. Check the body is physical rather than static.
A large scene runs slowly	Try Options > Video > Min, then Save settings to keep it. Physics > Simulation > Min-spec physics selects 60 Hz and four iterations; this remembered choice can make complex stacks less stable.
Clicks keep adding objects	Choose Select [Q] before selecting an existing object.
The cursor disappears	Release right mouse. A placement preview can replace the normal cursor over the world.
No placement preview	Choose an object type and move over terrain inside the viewport, clear of panels.
Shortcuts do nothing	Finish editing the text field or close the popup. Check the mode and release right mouse for tool shortcuts.
Controls are missing	Scroll inside panels or open their docks. Collapse the tutorial with its minus button. Press 0 if the UI is hidden.
My scene is missing after restart	New scene alone does not save it. Use Editor > Controls > Save scene and check the save feedback.
Files do not save or assets are missing	Extract the whole ZIP to a writable folder and use Launch SkullbonezCore.cmd. Do not run from inside the ZIP.
A tutorial step is stuck	Use Retry section for a fresh chapter setup. Resume restores a chapter, not a previous session's temporary recording.

Terrain and water

In Edit mode, open **Terrain**, enable **Terrain brush**, then hold **left mouse** to paint. Use **Brush: Raise ground / Lower ground** to choose the direction and **Brush radius** to set its size. Each brush drag supports **Ctrl+Z / Ctrl+Y** undo/redo. Right-drag moves the camera. Use **Water level** and **Show water** for a shoreline. **Save scene** keeps the terrain.

A useful first practice session

Create a new scene, place one static box and one dynamic ball, save, leave editing and watch. Reload, move the ball higher, save again and use **Predict** to explore the result. Or follow **Tutorial** from its camera lesson through the Solar System finale.

About this guide

A practical guide to **SkullbonezCore 2026.09.27**, covering the editor, camera navigation, scene saving, Replay and tutorial controls a new user needs first.

[Downloads and feedback](#) - include your build version, what you did and what happened when reporting a problem. The online **LLM Level-Authoring Guide** is linked from the repository README.